

## 5. Customer life value (CLV)

### 5.1 Aljabar limit

#### a. Hitung limit

$$L_1 = \lim_{u \rightarrow k} \frac{u^2 - k^2}{u - k} \Rightarrow \lim_{u \rightarrow k} \frac{(u-k)(u+k)}{u-k} = u+k$$

Substitusi  $u=k$

$$= 2k = 16$$

$$L_2 = \lim_{u \rightarrow 0} \frac{\sqrt{u+(u+1)^2} - (u+1)}{u} \Rightarrow \frac{u}{u(\sqrt{u+(u+1)^2} + (u+1))} \Rightarrow \frac{1}{\sqrt{u+(u+1)^2} + (u+1)}$$

Substitusi  $u$

$$= \frac{1}{2(u+1)} = \frac{1}{18}$$

$$L_3 = \lim_{u \rightarrow \infty} \frac{(a+2)u^3 + bu + 7}{u^3 - u + k} \xrightarrow{\text{Ambil suku tertinggi}} \frac{a+2}{1} = 3$$

### 5.2 Limit trigonometri

$$= \lim_{u \rightarrow 0} \frac{\sin((a+1)u) \tan((b+2)u)}{u^2}$$

$$\text{Gunakan } \sin u \sim u, \tan u \sim u$$

Jadi

$$\frac{(a+1)u \cdot (b+2)u}{u^2} = (a+1)(b+2) \Rightarrow 2 \cdot 8 = 16$$

### 5.3 CLV dan limit aljabar

$$D. R(n) = \frac{100n + k}{n + (a+b+1)} \quad \text{Limit} = \frac{100}{1} = 100$$

$$= \frac{100n + d}{n + 0} \quad \text{jadi, Steady state nya } 100 \text{ juta}$$

$$e. \lim_{n \rightarrow \infty} \left[ \sqrt{n^2 + kn} - \sqrt{n^2 + bn} \right] = \frac{0-0}{2} = 1$$

$$= \frac{(n^2 + kn) - (n^2 + bn)}{n(\sqrt{1+k/n} + \sqrt{1+b/n})}$$

$$= k-b/2$$



#### 5.4 (f) Kontinuitas

$$f(u) = \begin{cases} u^2 - u^2/u - k, & u < k \\ p, & u = k \\ au + q, & u > k \end{cases}$$

Syarat kontinu

1. Limit kiri =  $u + k \Rightarrow 2k = 16$

2. Nilai tengah =  $p = 16$

3. Limit kanan =  $ak + q = 8 + q$

diskontinuitas =  $8 + q \neq 16 \Rightarrow q \neq 8$

(u) > 0 maka nilai kontinuitas  
 limit kiri = 16  
 limit kanan = 16  
 nilai tengah = 16  
 maka kontinuitas = 16

$N = 16$ , kontinuitas  
 $16 = 16$

$(1+N) - (1+N) + 16$ , nilai = 16

$\frac{1}{16} - \frac{1}{16} = 0$

limit kanan =  $\frac{1}{16}$

limit kiri =  $\frac{1}{16}$

limit =  $\frac{1}{16}$

$16 = 8 \cdot 2 \Rightarrow (1+2)(1+2) = 16$

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